

1

Def: Symmetrization Let $p : R^N \rightarrow R$ be a polynomial, then $L(p)(x) = \frac{1}{N!} \sum_{\pi \in S_n} p(x \circ \pi)$ is the symmetrization of p

2

Basic Symmetric example: If p is already symmetric, then symmetrization should do nothing. Let p be given as

$$p(x_1, x_2) = x_1^2 + x_2^2$$

Then,

$S_2 = \{\pi_1, \pi_2\}$ since there are $2!$ unique permutations on tuples of 2 elements. So the symmetrization of p is given by

$$\begin{aligned} L(p) &= \frac{1}{2!} \sum_{\pi \in S_2} p(x_{\pi(1)}, x_{\pi(2)}) = \frac{1}{2} [p(x_{\pi_1(1)}, x_{\pi_1(2)}) + p(x_{\pi_2(1)}, x_{\pi_2(2)})] \\ &= \frac{1}{2} [p(y, x) + p(x, y)] \\ &= \frac{1}{2} [(y^2 + x^2) + (x^2 + y^2)] \\ &= \frac{1}{2} [y^2 + x^2 + x^2 + y^2] \\ &= x^2 + y^2 \end{aligned}$$

Which is exactly equal to p

3

Symmetrization of non-symmetric. Let p be given by

$$p(x, y) = x^2 + y - 1$$

Then $S_2 = \{\pi_1, \pi_2\}$ and

$$\begin{aligned} L(p) &= \frac{1}{2!} \sum_{\pi_k \in S_2} p(\pi_k(x), \pi_k(y)) \\ &= \frac{1}{2} [p(\pi_1(x), \pi_1(y)) + p(\pi_2(x), \pi_2(y))] \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} [p(y, x) + p(x, y)] \\
&= \frac{1}{2} [y^2 + x - 1 + x^2 + y - 1] \\
&= \frac{1}{2} [y^2 + x^2 + x + y - 2]
\end{aligned}$$

So,

$$p^{symm}(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{2}x + \frac{1}{2}y - 1$$

Which indeed is symmetric.

4: Theorem 1

We can generalize examples 2 and 3 to the following theorems (sect. 4 and 5)

Theorem: 1 Let $p \in R[x_1, \dots, x_n]$ be a multivariate polynomial in n variables. Then if p is symmetric, it shall be invariant under $L(p)$. Formally

$$\forall p \in R[x_1, \dots, x_n]: \quad isSymm(p) \implies p = p^{symm}, p^{symm} = L(p)$$

Proof:

Let $p(x_1, \dots, x_n)$ be an n variate polynomial. Suppose p is symmetric. This means that $\forall \pi_k \in S_n$, we have $p(x_{\pi_k(1)}, x_{\pi_k(2)}, \dots, x_{\pi_k(n)}) = p(x_1, x_2, \dots, x_n)$.

We want to show that p is invariant under L . That is $L(p(x_1, \dots, x_n)) = p(x_1, \dots, x_n)$.

Observe,

$$L(p(x_1, \dots, x_n)) = \frac{1}{n!} \sum_{\pi_k \in S_n} p(x_{\pi_k(1)}, \dots, x_{\pi_k(n)})$$

p is symmetric, so each term in the sum is exactly to p

$$L(p(x_1, \dots, x_n)) = \frac{1}{n!} \sum_{\pi_k \in S_n} p(x_1, \dots, x_n)$$

The set of permutation maps on n elements is $|S_n| = n!$ and each term of the sum is equal.

$$L(p(x_1, \dots, x_n)) = \frac{1}{n!} n! p(x_1, \dots, x_n)$$

$$L(p(x_1, \dots, x_n)) = p(x_1, \dots, x_n)$$

This gives the claim. $\square$

5: Theorem 2

Theorem: 2 Let $p \in R[x_1, \dots, x_n]$ be a multivariate polynomial in n variables. Then if p is not symmetric, then certainly it shall be symmetric on application of L . Formally, the following is a true statement

$$\forall p \in R[x_1, \dots, x_n]: \quad isSymmetric(L(p))$$

Proof:

Let $p \in R[x_1, \dots, x_n]$. We want to show p under L is symmetric. This means that $\forall \pi_k \in S_n$, we have $L(p)(x_{\pi_k(1)}, x_{\pi_k(2)} \dots, x_{\pi_k(n)}) = L(p)(x_1, x_2, \dots, x_n)$.

Observe, we can consider two cases. p is either already symmetric or it is not.

Case 1: p is symmetric

From theorem 1, we have the claim

Case 2: p is not symmetric.

Let $\tau \in S_n$ be particular permutation on the set $[n]$. Then we want to show

$$L(p)(x_{\tau(1)}, x_{\tau(2)} \dots, x_{\tau(n)}) = L(p)(x_1, x_2, \dots, x_n)$$

Using the definition of L

$$L(p(x_1 \circ \tau, \dots, x_n \circ \tau)) = \frac{1}{n!} \sum_{\pi_k \in S_n} p(x \circ \tau \circ \pi_k, \dots, x \circ \tau \circ \pi_k)$$

The summation is over S_n , the set of permutations on the set $[n]$. By assumption, $\tau \in S_n$. So, we can split the sum as follows

$$L(p(x_1 \circ \tau, \dots, x_n \circ \tau)) = \frac{1}{n!} p(x \circ \tau \circ \tau, \dots, x \circ \tau \circ \tau) + \frac{1}{n!} \sum_{\pi_k \in (S_n / \tau)} p(x \circ \tau \circ \pi_k, \dots, x \circ \tau \circ \pi_k)$$

Clearly, $\tau \circ \tau = I$

$$L(p(x_{\tau(1)}, \dots, x_{\tau(n)})) = \frac{1}{n!} p(x_1, \dots, x_n) + \frac{1}{n!} \sum_{\pi_k \in S_n} p(x_{\pi_k(\tau(1))}, \dots, x_{\pi_k(\tau(n))})$$

For a fixed term in the summation, we can write $\pi' = \pi_k \circ \tau$. So $\pi' \in S_n$. S_n contains all permutations on $[n]$, except the identity permutation. Therefore, we can add back the split term to get

$$L(p(x_{\tau(1)}, \dots, x_{\tau(n)})) = \frac{1}{n!} \sum_{\pi'_k \in S_n} p(x_{\pi'_k(\tau(1))}, \dots, x_{\pi'_k(\tau(n))}) = L(p(x_1, \dots, x_n))$$

This gives the claim. $\square$